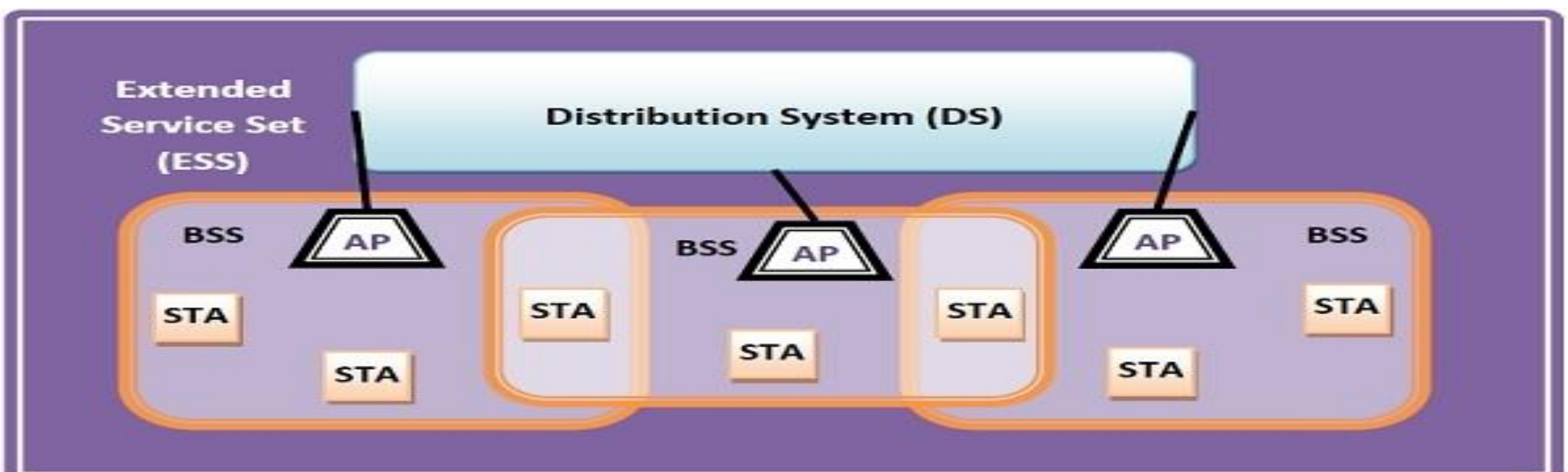
MODULE V

IEEE 802.11 Wi-Fi, Bluetooth, Threats and attacks, Network Address Translation, Firewalls, VPNs, Introduction to Network Management, SNMP.

IEEE 802.11 (Wi-Fi)

- IEEE 802.11 standard, popularly known as WiFi, lays down the architecture and specifications of wireless LANs (WLANs).
- The Institute of Electrical and Electronics Engineers (IEEE) uses its platform for developing technology standards
- WiFi or WLAN uses high-frequency radio waves instead of cables for connecting the devices in LAN.
- Users connected by WLANs can move around within the area of network coverage.

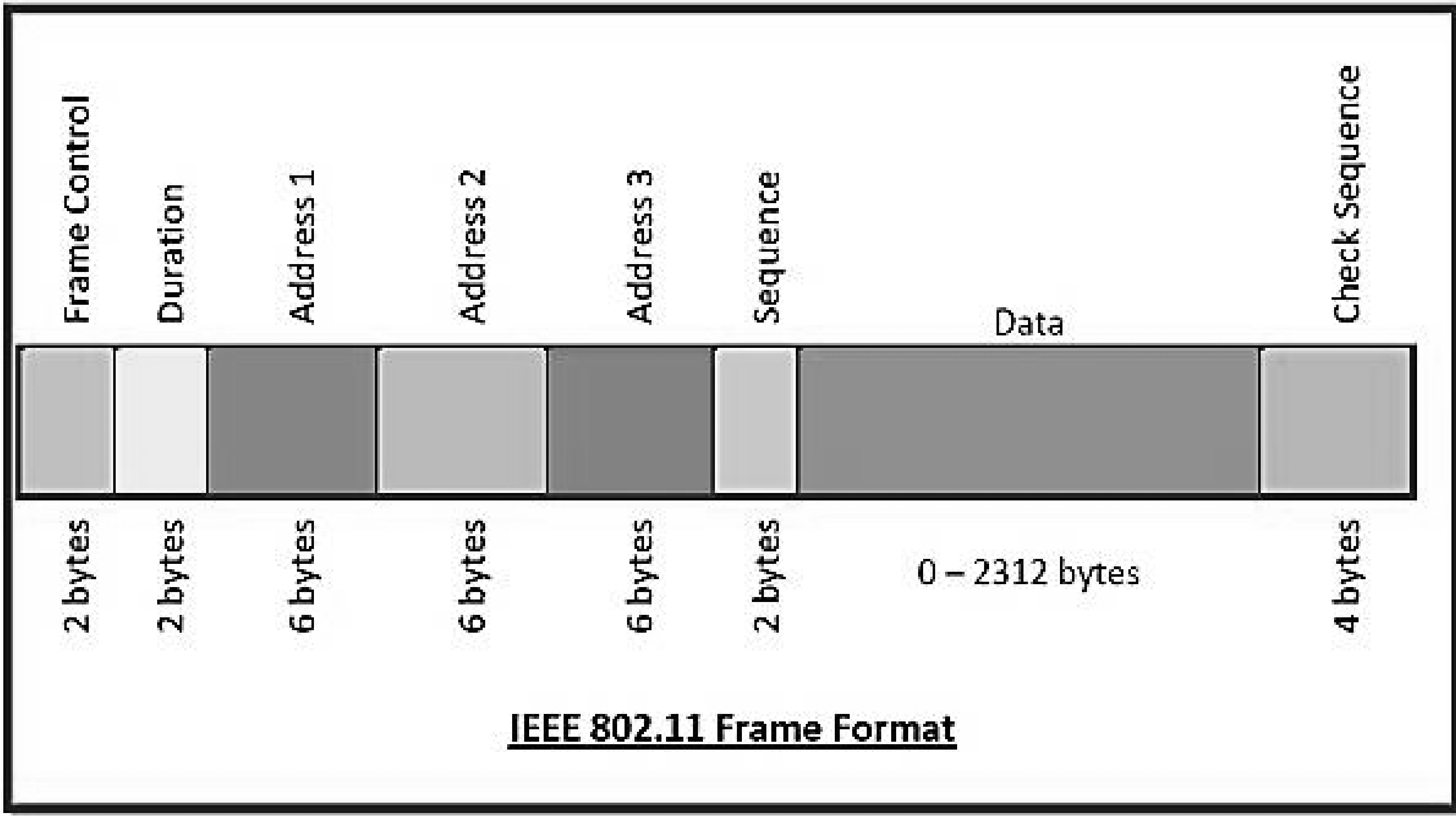
IEEE 802.11 ARCHITECTURE



The components of an IEEE 802.11 architecture are as follows −

- **Stations (STA)** − Stations comprises of **all devices and equipment** that are connected to the wireless LAN. A station can be of two types−
  - **Wireless Access Point (WAP)** − WAPs or simply access points (AP) are generally **wireless routers** that form the base stations or access.
  - **Client-** Clients are workstations, computers, laptops, printers, smartphones, etc.
- Each station has a **wireless network interface controller.**
- **Basic Service Set (BSS)** − A basic service set is a **group of stations** communicating at the physical layer level. BSS can be of two categories depending upon the mode of operation−
  - **Infrastructure BSS** − Here, the devices communicate with other devices through access points.
  - **Independent BSS −** Here, the devices communicate in a peer-to-peer basis in an ad hoc manner.
- **Extended Service Set (ESS)** − It is a set of all connected BSS.
- **Distribution System (DS)** − It connects access points in ESS.

Frame Format of IEEE 802.11



The main fields of a frame of wireless LANs as laid down by IEEE 802.11 are −

- **Frame Control** − It is a 2 bytes starting field composed of 11 subfields. It contains **control information** of the frame.
- **Duration** − It is a 2-byte field that specifies the **time period** for which the frame and its acknowledgment occupy the channel.
- **Address fields** − There are three 6-byte address fields containing addresses of source, destination
- **Sequence** − It a 2 bytes field that stores the frame numbers.
- **Data** − This is a variable-sized field that carries the data from the upper layers. The maximum size of the data field is 2312 bytes.
- **Check Sequence** − It is a 4-byte field containing error detection information.

ADVANTAGES	DISADVANTAGES
• The IEEE 802.11 is easy to installation. • It has higher frequency range. • It has efficient coding technique. • The IEEE 802.11 has reduced wiring expense. • It is less cost effective.	• It has traffic disruptions. • Network security and the maintenance needed to stay secured. • It is required periodic maintenance. • Data transmitted over radio waves can be captured by any Wi-Fi ready devices in the area. • It has unauthorized uses. • Frame spoofing. • Session hijacking.

BLUETOOTH

- Bluetooth is universal for short-range wireless voice and data communication.
- It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances.
- This technology was invented by Ericson in 1994.
- It operates in the unlicensed, industrial, scientific, and medical (ISM) band from 2.4 GHz to 2.485 GHz.
- Maximum devices that can be connected at the same time are 7.
- Bluetooth ranges up to 10 meters.
- It provides data rates up to 1 Mbps or 3 Mbps depending upon the version.
- The spreading technique that it uses is FHSS (Frequency-hopping spread spectrum).
- A Bluetooth network is called a **piconet** and a collection of interconnected piconets is called **scatternet**.

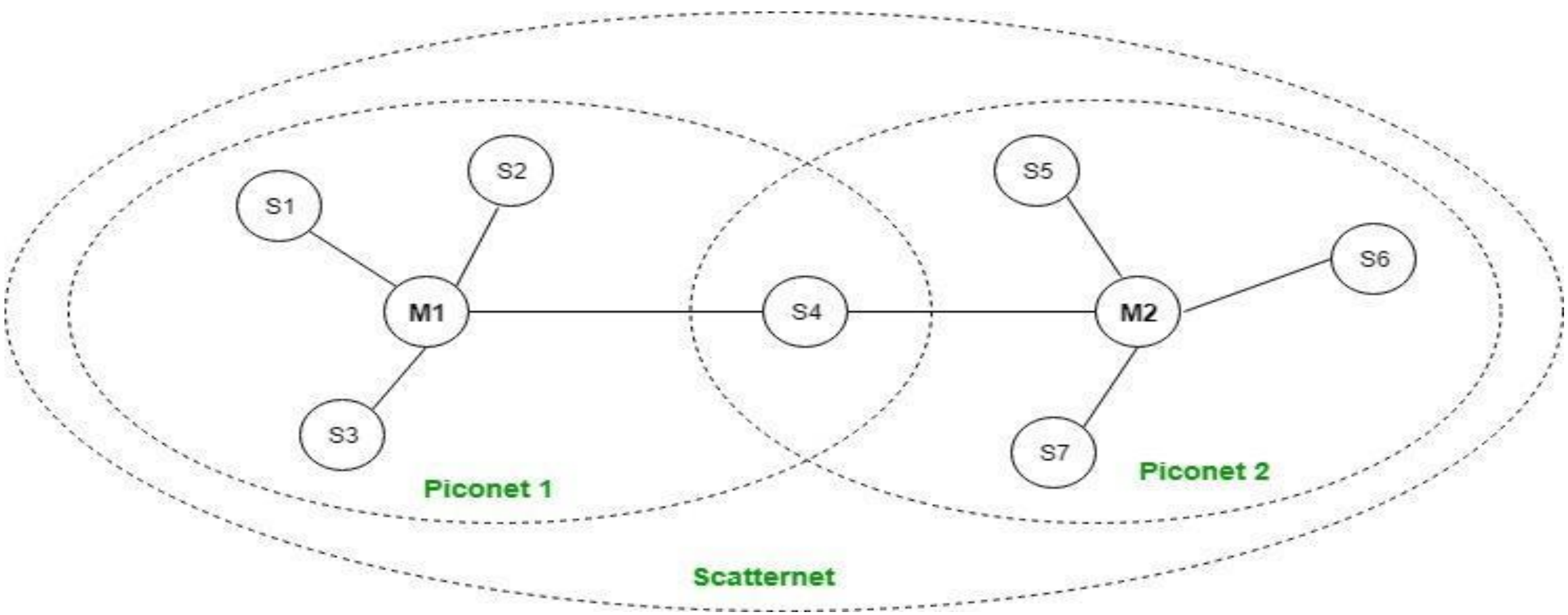
What is Bluetooth?

- Bluetooth simply follows the principle of transmitting and receiving data using radio waves.
- It can be paired with the other device which has also Bluetooth but it should be within the estimated communication range to connect.
- When two devices start to share data, they form a network called piconet which can further accommodate more than five devices.
- Bluetooth Transmission capacity 720 kbps.
- Bluetooth is Wireless.
- Bluetooth is a Low-cost short-distance radio communications standard.
- Bluetooth is robust and flexible.
- Bluetooth is cable replacement technology that can be used to connect almost any device to any other device.
- The basic architecture unit of Bluetooth is a piconet.

Bluetooth Architecture:

The architecture of Bluetooth defines two types of networks:

1. Piconet
2. Scatternet



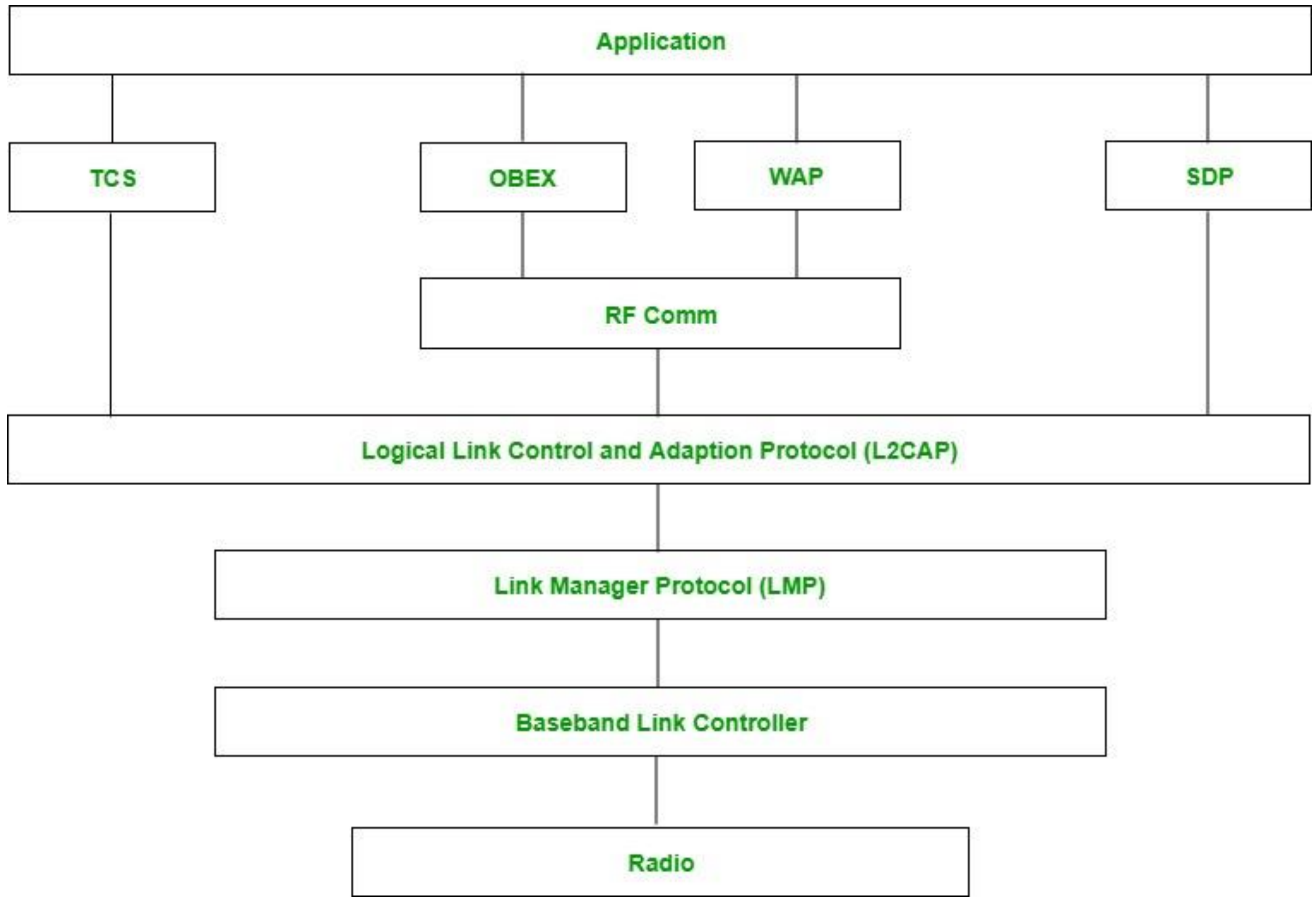
Piconet:

- Piconet is a type of Bluetooth network that contains **one primary node** called the master node and **seven active secondary nodes** called slave nodes.
- Thus, we can say that there is a total of 8 active nodes which are present at a distance of 10 meters.
- The communication between the primary and secondary nodes can be one-to-one or one-to-many.
- Possible communication is only between the master and slave; Slave-slave communication is not possible.
- It also has **255 parked nodes**, these are secondary nodes and cannot take participation in communication unless it gets converted to the active state.

Scatternet:

- It is formed by using **various piconets**.
- A slave that is present in one piconet can act as master or we can say primary in another piconet.
- This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master.
- This type of node is referred to as a bridge node.
- A station cannot be mastered in two piconets.

Bluetooth protocol stack:



- **Radio (RF) layer:** It specifies the details of the air interface, including frequency, the use of frequency hopping and transmit power. It performs modulation/demodulation of the data into RF signals. It defines the physical characteristics of Bluetooth transceivers. It defines two types of physical links: connection-less and connection-oriented.
- **Baseband Link layer:** The baseband is the digital engine of a Bluetooth system and is equivalent to the MAC sublayer in LANs. It performs the connection establishment within a piconet, addressing, packet format, timing and power control.
- **Link Manager protocol layer:** It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
- **Logical Link Control and Adaption (L2CAP) Protocol layer:** It is also known as the heart of the Bluetooth protocol stack. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.

- **Service Discovery Protocol (SDP) layer:** It is short for Service Discovery Protocol. It allows discovering the services available on another Bluetooth-enabled device.
- **RF comm layer:** It is a cabal replacement protocol. It is short for Radio Frontend Component. It provides a serial interface with WAP and OBEX. It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP). The protocol is based on the ETSI standard TS 07.10.
- **OBEX:** It is short for Object Exchange. It is a communication protocol to exchange objects between 2 devices.
- **WAP:** It is short for Wireless Access Protocol. It is used for internet access.
- **TCS:** It is short for Telephony Control Protocol. It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
- **Application layer:** It enables the user to interact with the application.

Types of Bluetooth

Various types of Bluetooth are available in the market nowadays.

- **In-Car Headset**: One can make calls from the car speaker system without the use of mobile phones.
- **Stereo Headset**: To listen to music in car or in music players at home.
- **Webcam**: One can link the camera with the help of Bluetooth with their laptop or phone.
- **Bluetooth-equipped Printer**: The printer can be used when connected via Bluetooth with mobile phone or laptop.
- **Bluetooth Global Positioning System (GPS)**: To use GPS in cars, one can connect their phone with car system via Bluetooth to fetch the directions of the address.

ADVANTAGES	DISADVANTAGES
• It is a low-cost and easy-to-use device. • It can also penetrate through walls. • It creates an Ad-hoc connection immediately without any wires. • It is used for voice and data transfer.	• It can be hacked and hence, less secure. • It has a slow data transfer rate: of 3 Mbps. • It has a small range: 10 meters. • Bluetooth communication does not support routing. • The issues of handoffs have not been addressed.

Applications:

- It can be used in laptops, and in wireless PCs, printers.
- It can be used in wireless headsets, wireless PANs, and LANs.
- It can connect a digital camera wirelessly to a mobile phone.
- It can transfer data in terms of videos, songs, photographs, or files from one cell phone to another cell phone or computer.
- It is used in the sectors of Medical health care, sports and fitness, Military.

NETWORK ATTACK AND THREATS

- A network attack is an attempt to gain unauthorized access to an organization's network, with the objective of stealing data or perform other malicious activity. There are two main types of network attacks:
- **Passive:** Attackers gain access to a network and can monitor or steal sensitive information, but without making any change to the data, leaving it intact.
- **Active:** Attackers not only gain unauthorized access but also modify data, either deleting, encrypting or otherwise harming it.

We distinguish network attacks from several other types of attacks:

- **Endpoint attacks**—gaining unauthorized access to user devices, servers or other endpoints, typically compromising them by infecting them with malware.
- **Malware attacks**—infecting IT resources with malware, allowing attackers to compromise systems, steal data and do damage. These also include ransomware attacks.
- **Vulnerabilities, exploits and attacks**—exploiting vulnerabilities in software used in the organization, to gain unauthorized access, compromise or sabotage systems.
- **Advanced persistent threats**—these are complex multilayered threats, which include network attacks but also other attack types.

What are the Common Types of Network Attacks?

Following are common threat vectors attackers can use to penetrate your network.

1. Unauthorized access : Unauthorized access refers to attackers accessing a network without receiving permission. Among the causes of unauthorized access attacks are weak passwords, lacking protection against social engineering, previously compromised accounts, and insider threats.

2. Distributed Denial of Service (DDoS) attacks : Attackers build botnets, large fleets of compromised devices, and use them to direct false traffic at your network or servers. DDoS can occur at the network level, for example by sending huge volumes of SYN/ACC packets which can overwhelm a server, or at the application level, for example by performing complex SQL queries that bring a database to its knees.

3. Man in the middle attacks: A man in the middle attack involves attackers intercepting traffic, either between your network and external sites or within your network. If communication protocols are not secured or attackers find a way to circumvent that security, they can steal data that is being transmitted, obtain user credentials and hijack their sessions.

4. Code and SQL injection attacks: Many websites accept user inputs and fail to validate and sanitize those inputs. Attackers can then fill out a form or make an API call, passing malicious code instead of the expected data values. The code is executed on the server and allows attackers to compromise it.

5. Privilege escalation: Once attackers penetrate your network, they can use privilege escalation to expand their reach. Horizontal privilege escalation involves attackers gaining access to additional, adjacent systems, and vertical escalation means attackers gain a higher level of privileges for the same systems.

6. Insider threats:A network is especially vulnerable to malicious insiders, who already have privileged access to organizational systems. Insider threats can be difficult to detect and protect against, because insiders do not need to penetrate the network in order to do harm. New technologies like User and Even Behavioral Analytics (UEBA) can help identify suspicious or anomalous behavior by internal users, which can help identify insider attacks.

Network Protection Best Practices

- **Segregate Your Network**: A basic part of avoiding network security threats is dividing a network into zones based on security requirements. This can be done using subnets within the same network, or by creating Virtual Local Area Networks (VLANs), each of which behaves like a complete separate network. Segmentation limits the potential impact of an attack to one zone, and requires attackers to take special measures to penetrate and gain access to other network zones.

- **Regulate Access to the Internet via Proxy Server**: Do not allow network users to access the Internet unchecked. Pass all requests through a transparent proxy, and use it to control and monitor user behavior. Ensure that outbound connections are actually performed by a human and not a bot or other automated mechanism. Whitelist domains to ensure corporate users can only access websites you have explicitly approved.

- **Place Security Devices Correctly**: Place a firewall at every junction of network zones, not just at the network edge. If you can't deploy full-fledged firewalls everywhere, use the built-in firewall functionality of your switches and routers. Deploy anti-DDoS devices or cloud services at the network edge. Carefully consider where to place strategic devices like load balancers – if they are outside the Demilitarized Zone (DMZ), they won't be protected by your network security apparatus.

- **Use Network Address Translation**: Network Address Translation (NAT) lets you translate internal IP addresses into addresses accessible on public networks. You can use it to connect multiple computers to the Internet using a single IP address. This provides an extra layer of security, because any inbound or outgoing traffic has to go through a NAT device, and there are fewer IP addresses which makes it difficult for attackers to understand which host they are connecting to.
- **Monitor Network Traffic**: Ensure you have complete visibility of incoming, outgoing and internal network traffic, with the ability to automatically detect threats, and understand their context and impact. Combine data from different security tools to get a clear picture of what is happening on the network, recognizing that many attacks span multiple IT systems, user accounts and threat vectors.
- Achieving this level of visibility can be difficult with traditional security tools. Cynet 360 is an integrated security solution offering advanced network analytlics which continuously monitors network traffic, automatically detect malicious activity, and either respond to it automatically or pass context-rich information to security staff.
- **Use Deception Technology**: No network protection measures are 100% successful, and attackers will eventually succeed in penetrating your network. Recognize this and place deception technology in place, which creates decoys across your network, tempting attackers to “attack” them, and letting you observe their plans and techniques. You can use decoys to detect threats in all stages of the attack lifecycle: data files, credentials and network connections.
- Cynet 360 is an integrated security solution with built-in deception technology which provides both off-the-shelf decoy files and the ability to create decoys to meet your specific security needs. , while taking into account your environment’s security needs.

NETWORK ADDRESS TRANSLATION (NAT)

- NAT (Network Address Translation) is a technique used in computer networks to modify IP address information in packet headers as they pass through a router or firewall.
- It allows multiple devices on a private network to share a single public IP address.

Why NAT is used

- Conserves limited public IPv4 addresses
- Enhances basic security (internal IPs are hidden)
- Allows private networks to access the internet
- Without NAT, every device would need a unique public IP (impractical)

Imagine a company office:

- Employees inside the office = **devices with private IP addresses**
- Office main phone number = **public IP address**
- Receptionist = **NAT router**

Step 1: Employee Makes a Call

- An employee wants to call someone outside:
- They don’t use their personal number
- They use the **office’s main number**
- (Private IP → Public IP)

Step 2: Receptionist Keeps Record

- Receptionist notes:
  - Who made the call
  - Which extension (like port number)
- (This is like the **NAT table**)

Step 3: Response Comes Back

- The outside person calls back on the same office number
- Receptionist checks records
- Transfers call to correct employee
- (Public IP → Private IP)

Network Address Translation (NAT) working

Network Address Translation (NAT) is typically configured on a **border router/Firewall**, which connects a local (inside) network to a global (outside) network (Internet).

The router has:

- One inside interface (connected to private network)
- One outside interface (connected to public network)

- **Outgoing Packet (Inside → Outside)**

When a packet leaves the local network:

  - NAT converts the **private (local) IP address** into a **public (global) IP address**
  - It may also modify the **port number**
  - A mapping entry is created in the **NAT table**

- **Incoming Packet (Outside → Inside)**

When a packet enters the local network:

  - NAT checks the **NAT table**
  - Converts the **public IP address back to the private IP address**
  - Forwards the packet to the correct internal host

Address Pool Exhaustion

- If NAT uses a **pool of public IP addresses** and all are in use:
  - New packets cannot be translated
  - Such packets are **dropped**
  - An **ICMP "Host Unreachable" message** may be sent to the sender

Why mask port numbers ?

- In a network, multiple hosts may try to communicate with the same destination server using the same port number at the same time.
- If NAT only translates IP addresses, then all packets would appear to come from the **same public IP address with the same port number**, making it impossible for the NAT device to identify which response belongs to which host.
- Therefore, NAT **masks (translates) the port numbers** as well, assigning a unique port number to each connection. This ensures that return packets are correctly mapped to the respective host using the NAT table

Problem Without Port Masking

Suppose:

- Host A → 192.168.1.2:1000
- Host B → 192.168.1.3:1000

Both send requests to the same destination

If NAT only translates **IP addresses**:

Both packets will appear as coming from the **same public IP**

Source port for both = 1000

⍰ So packets look like:

- 203.0.113.5:1000 → Destination
  203.0.113.5:1000 → Destination

Issue

When the destination replies:

Reply comes back to:

203.0.113.5:1000

⍰ NAT **cannot distinguish**:

- Which packet belongs to Host A
- Which packet belongs to Host B

Solution: Port Masking (PAT)

To solve this, NAT also translates **port numbers**:

- Host A → 203.0.113.5:1001
- Host B → 203.0.113.5:1002

Now packets are:

- 203.0.113.5:1001 → Destination
  203.0.113.5:1002 → Destination

Result

- Replies return to different ports
- NAT uses the **NAT table** to map them back correctly

Private IP	Private Port	Public IP	Public Port
192.168.1.2	1000	203.0.113.5	1001
192.168.1.3	1000	203.0.113.5	1002

Network Address Translation (NAT) Types

There are 3 ways to configure NAT:

Static NAT

One-to-one mapping between private and public IP

Used for hosting servers

Example:

Private IP 192.168.1.10 → Public IP 203.0.113.5.

Suppose, if there are 3000 devices that need access to the Internet, the organization has to buy 3000 public addresses that will be very costly.

Dynamic NAT

Maps private IPs to a pool of public IPs

Assigned dynamically when needed

Suppose, if there is a pool of 2 public IP addresses then only 2 private IP addresses can be translated at a given time. If 3rd private IP address wants to access the Internet then the packet will be dropped therefore many private IP addresses are mapped to a pool of public IP addresses. NAT is used when the number of users who want to access the Internet is fixed. This is also very costly as the organization has to buy many global IP addresses to make a pool.

Port Address Translation (PAT)

Most common type

Multiple devices share one public IP using different port numbers

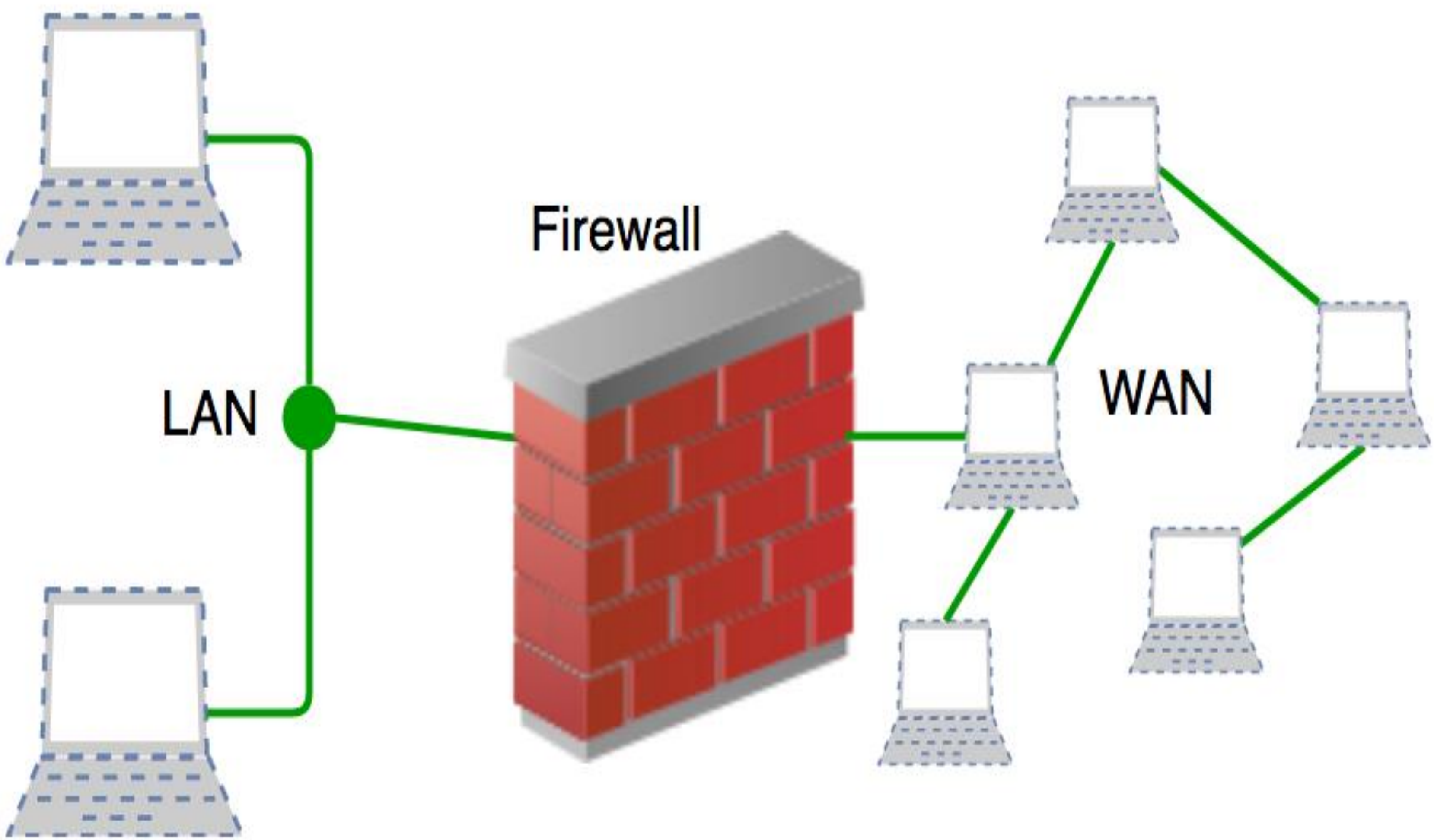
Example: 192.168.1.2:5000 → 203.0.113.5:1001

192.168.1.3:5001 → 203.0.113.5:1002

FIREWALL

Working of a Firewall

- **Traffic Inspection:** Checks all incoming and outgoing packets.
- **Filtering:** Compares packets against **security rules**.
- **Decision:**
  - Allow packet → forwards to network
  - Block packet → drops or logs it
- **Logging & Alerts:** Records suspicious activity for monitoring and auditing.


Firewall
LAN
WAN

History and Need for Firewall

- Before Firewalls, network security was performed by Access Control Lists (ACLs) residing on routers.
- ACLs are rules that determine whether network access should be granted or denied to specific IP address.
- But ACLs cannot determine the nature of the packet it is blocking. Also, ACL alone does not have the capacity to keep threats out of the network.
- Hence, the Firewall was introduced. Connectivity to the Internet is no longer optional for organizations.
- However, accessing the Internet provides benefits to the organization; it also enables the outside world to interact with the internal network of the organization.
- This creates a threat to the organization. In order to secure the internal network from unauthorized traffic, we need a Firewall.

Generations of firewalls:

First-Generation Firewalls (1980s-1990s)

- Packet Filtering Firewalls: These firewalls examine packet headers to filter traffic based on source and destination IP addresses, ports, and protocols.
- Characteristics: Simple, fast, and inexpensive.
- Limitations: Do not examine packet contents, vulnerable to spoofing attacks.

Second Generation Firewalls (1990s 2000s)

- Stateful Inspection Firewalls: These firewalls examine packet headers and contents to filter traffic based on the context of the connection.
- Characteristics: More secure than packet filtering firewalls, can detect and prevent spoofing attacks.
- Limitations: Can be slower than packet filtering firewalls, may require more configuration.

Third Generation Firewalls (2000s present)

- Application Layer Firewalls: These firewalls examine packet contents at the application layer to filter traffic based on specific applications or services.
- Characteristics: Can detect and prevent attacks that exploit application vulnerabilities, provide more granular control over network traffic.
- Limitations: Can be complex to configure, may require significant resources.

Fourth Generation Firewalls (2010s present)

- Next Generation Firewalls (NGFWs) : These firewalls combine the features of traditional firewalls with advanced security features, such as intrusion prevention, antivirus, and sandboxing.
- Characteristics : Provide comprehensive security features, can detect and prevent advanced threats, provide visibility into network traffic.
- Limitations : Can be complex to configure, may require significant resources.

Fifth Generation Firewalls (2020s present)

- Cloud Native Firewalls : These firewalls are designed specifically for cloud environments and provide advanced security features, such as micro segmentation and serverless architecture.
- Characteristics : Provide scalable and flexible security for cloud environments, can detect and prevent advanced threats, provide visibility into network traffic.
- Limitations : Still evolving, may require significant resources and expertise to configure and manage.

Types of Firewall

Firewalls are generally of two types: *Host-based* and *Network-based.*

- **Host- based Firewalls :** Host-based firewall is installed on each network node which controls each incoming and outgoing packet. It is a software application or suite of applications, comes as a part of the operating system. Host-based firewalls are needed because network firewalls cannot provide protection inside a trusted network. Host firewall protects each host from attacks and unauthorized access.
- **Network-based Firewalls :** Network firewall function on network level. In other words, these firewalls filter all incoming and outgoing traffic across the network. It protects the internal network by filtering the traffic using rules defined on the firewall. A Network firewall might have two or more network interface cards (NICs). A network-based firewall is usually a dedicated system with proprietary software installed.

Advantages of using Firewall

- **Protection from unauthorized access:** Firewalls can be set up to restrict incoming traffic from particular IP addresses or networks, preventing hackers or other malicious actors from easily accessing a network or system. Protection from unwanted access.
- **Prevention of malware and other threats:** Malware and other threat prevention: Firewalls can be set up to block traffic linked to known malware or other security concerns, assisting in the defense against these kinds of attacks.
- **Control of network access:** By limiting access to specified individuals or groups for particular servers or applications, firewalls can be used to restrict access to particular network resources or services.
- **Monitoring of network activity:** Firewalls can be set up to record and keep track of all network activity. This information is essential for identifying and looking into security problems and other kinds of shady behavior.
- **Regulation compliance:** Many industries are bound by rules that demand the usage of firewalls or other security measures. Organizations can comply with these rules and prevent any fines or penalties by using a firewall.
- **Network segmentation:** By using firewalls to split up a bigger network into smaller subnets, the attack surface is reduced and the security level is raised.

Disadvantages of using Firewall

- **Complexity:** Setting up and keeping up a firewall can be time-consuming and difficult, especially for bigger networks or companies with a wide variety of users and devices.
- **Limited Visibility:** Firewalls may not be able to identify or stop security risks that operate at other levels, such as the application or endpoint level, because they can only observe and manage traffic at the network level.
- **False sense of security:** Some businesses may place an excessive amount of reliance on their firewall and disregard other crucial security measures like endpoint security or intrusion detection systems.
- **Limited adaptability:** Because firewalls are frequently rule-based, they might not be able to respond to fresh security threats.
- **Performance impact:** Network performance can be significantly impacted by firewalls, particularly if they are set up to analyze or manage a lot of traffic.
- **Limited scalability:** Because firewalls are only able to secure one network, businesses that have several networks must deploy many firewalls, which can be expensive.
- **Limited VPN support:** Some firewalls might not allow complex VPN features like split tunneling, which could restrict the experience of a remote worker.
- **Cost:** Purchasing many devices or add-on features for a firewall system can be expensive, especially for businesses.

Real-Time Applications of Firewall

- Corporate networks
- Government organizations
- Service providers
- Small enterprises
- Networks at home:
- Industrial Control Systems (ICS)

Examples

- Home Wi-Fi router firewall blocking incoming unsolicited connections.
- Corporate office firewall preventing employees from accessing social media websites.
- Data centres using NGFW (Next-Generation Firewall) to block malware and detect intrusion attempts.

<u>SNMP</u>

- SNMP stands for **Simple Network Management Protocol**. It is used to **monitor and manage network devices** like routers, switches, servers…
- It works at the **Application Layer** of the network.
- It is developed by the **Internet Engineering Task Force (IETF)**.

Why SNMP is Used

SNMP helps network administrators to:

- Check network performance
- Detect problems quickly
- Control and configure devices
- Collect information like CPU, memory, traffic
- Manage large networks easily

SNMP Architecture

- SNMP works using a **Manager–Agent model**.

a) SNMP Manager

- Central system (brain of network)
- Sends requests to devices
- Receives responses
- Example: Network monitoring software

b) SNMP Agent

- Software installed in devices
- Collects data from device
- Sends data to manager

c) Managed Devices

Devices that are monitored:

- Router
- Switch
- Server

d) MIB (Management Information Base)

- A **database** that stores information about devices
- Contains details like:
  - Device name
  - Speed
  - Status

e) OID (Object Identifier)

- Unique ID for each item in MIB
- Written in number format
- ? Example:
- 1.3.6.1.2.1

SNMP Operations

1. GET

- Manager asks for data
  Example: “What is CPU usage?”

2. GET-NEXT

- Gets next value in list

3. GET-BULK

- Gets large amount of data (faster)

4. SET

- Manager changes data
  Example: Change device setting

5. RESPONSE

- Agent replies to manager

6. TRAP

- Agent sends alert automatically
- Example:
- Link failure
- System crash

7. INFORM

- Same as TRAP but needs confirmation

How SNMP Works

SNMP communication happens between **Manager** and **Agent** using the **MIB database**.

Step 1: Manager Sends Request (GET / SET)

- The **SNMP Manager** (central system) starts the communication
- It sends a request to the **Agent** using SNMP protocol

Types of Requests:

- **GET** → To read data
- **SET** → To change data

Example:

- Manager asks: *“What is the CPU usage of router?”* (GET)
- Manager says: *“Change device name to Router1”* (SET)

This request contains:

- OID (what data is needed)
- Type of request
- Community string (like password)

Step 2: Agent Receives the Request

- The **Agent** is a software running inside the device (router/switch)
- It receives the request from manager

Now the agent:

- Checks if the request is valid
- Verifies permission using community string
- Identifies which data is needed

Step 3: Agent Checks MIB (Database Lookup)

- The agent searches the **MIB (Management Information Base)**
- ? MIB contains:
- All device information
- Each value stored with an **OID (Object Identifier)**

Example:

- OID → CPU usage
- OID → Device status
- Agent finds the exact value requested by manager

Step 4: Agent Sends Response Back

- After finding the data, the agent sends a **RESPONSE message**
- This response includes:
- Requested value (e.g., CPU = 40%)
- Status (success or error)
- Communication uses:
- **UDP Port 161**

Step 5: TRAP Message (If Problem Occurs)

- If any **error or event happens**, agent does NOT wait for manager
- It sends a **TRAP message automatically**
- ? TRAP is an **unsolicited message**

Example situations:

- Network link failure
- Device overheating
- System crash

? TRAP uses:

- **UDP Port 162**

VIRTUAL PRIVATE NETWORK

- A Virtual Private Network (VPN) is a technology that creates a secure and encrypted connection between a user's device and a VPN server.

What is a VPN?

- A VPN is a network that uses encryption and other security measures to protect data transmitted between a user's device and a VPN server. This creates a secure and private connection, even when using public Wi-Fi networks.

How Does a VPN Work?

Client Device:

- Can be a laptop, phone, or desktop.
- Connects to the VPN server using VPN software.

Encrypted Tunnel:

- All data is **encrypted** before leaving the device.
- Protects data from hackers and eavesdroppers.

VPN Server / Gateway:

- Decrypts the data.
- Forwards it to the destination server on the Internet.

Destination Server:

- The website or service the client wants to access.
- Replies go back through the **VPN server**, then encrypted back to the client

Issues Without VPN

Data is Not Encrypted

- Information is sent in **plain text** over the Internet.
- Hackers can easily **intercept passwords, emails, and sensitive data**.

IP Address is Exposed

- Your **real IP address** is visible to websites, ISPs, and potential attackers.
- Makes tracking your location and online activity easy.

Low Privacy

- Online activities can be **monitored by ISPs, government agencies, or malicious actors**.
- Browsing history and personal information can be collected.

Vulnerable on Public Wi-Fi

- Public networks (cafes, airports, hotels) are **easy targets for hackers**.
- Without a VPN, you can fall victim to **man-in-the-middle attacks**.

Cannot Bypass Geo-Restrictions

- Websites and streaming services may **block access based on location**.
- Without VPN, you can't access **region-restricted content**.

No Secure Remote Access

- Employees cannot safely **access corporate networks from home**.
- Increases the risk of **data breaches**.

Risk from Hackers & Malware

- Data packets are not encrypted → attackers can **inject malware, phishing attacks, or spyware**.

Limited Anonymity

- Websites can track you using:
  - IP address
  - Cookies
  - Browser fingerprinting

Types of VPNs

- **Remote Access VPN** : Allows users to connect to a remote network, such as a company's internal network.
- **Site-to-Site VPN** : Connects multiple networks, such as branch offices, to a central network.
- **Free VPN** : Offers limited VPN services for free, often with ads and limited bandwidth.
- **Paid VPN** : Offers premium VPN services with advanced features, faster speeds, and better security.

Common VPN Uses

- **Secure Public Wi-Fi** : VPNs protect data transmitted over public Wi-Fi networks.
- **Remote Work :** VPNs allow employees to securely connect to company networks.
- **Streaming and Gaming** : VPNs can improve streaming and gaming performance by reducing latency and packet loss.
- **Bypass Censorship** : VPNs can help users bypass internet censorship and restrictions.

Examples

- Employee works from home, accesses corporate network via VPN.
- A person watches region-restricted content using VPN.
- Connecting two branch offices securely over the Internet.